Eigenvalues and inverse problems

Deterministic regularization of the nonsymmetric eigenproblem

Difficulty: extreme **Importance:** broadly interesting

Status: open.

Problem statement

For a matrix M with simple spectrum, define

$$\operatorname{gap}(M) = \min_{i \neq j} |\lambda_i(M) - \lambda_j(M)|, \qquad \kappa_V(M) = \inf_{M = VDV^{-1},\, D \text{ diagonal}} \|V\|_2 \|V^{-1}\|_2.$$

For every $n \geq 2$, $A \in \mathbb{C}^{n \times n}$ with $\|A\|_2 \leq 1$, and $0 < \delta < 1/2$, construct deterministically a matrix E such that

$$\|E\|_2 \leq \delta, \qquad A + E \text{ has simple spectrum}, \qquad \frac{\kappa_V(A+E)}{\operatorname{gap}(A+E)} \leq C(n/\delta)^c,$$

using $O(n^3 \log^d(n/\delta))$ exact arithmetic operations, for universal constants C, c, d. The task is to find the perturbation, not merely prove its existence.

References

Banks, Garza-Vargas, Kulkarni, and Srivastava, *Pseudospectral Shattering, the Sign Function, and Diagonalization in Nearly Matrix Multiplication Time*, FOCM 23 (2023), §6, first future-research question. Amsel et al., *Linear Systems and Eigenvalue Problems*, August 2026, Problem 3.1.

Status check

Searches for *deterministic pseudospectral shattering 2026* found randomized and exponential-bound deterministic results, not the stated algorithm. The normalization and exponent orientation here are explicit: Problem 3.1's printed $(\delta/n)^c$ contradicts its own preceding motivation; $(n/\delta)^c$ is the intended polynomial upper bound.